

EMIRALI UZUN

CYBERSECURITY & SYSTEM ENGINEER

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PROFESSIONAL SUMMARY

Linux-focused software developer with strong hands-on experience in C, C++, system programming, Linux administration, and low-level software development. Experienced in process management, inter-process communication, multithreading, synchronization, memory management, file I/O, UNIX signals, shell development, networking, and Linux system configuration. Developed a Linux kernel camera driver for the NVIDIA Jetson Orin Nano Super, integrating V4L2, Device Tree, MIPI CSI-2, I2C, and TegraCam. Strong practical background in Linux environments including Debian, Red Hat, Arch Linux, and NixOS, with additional experience in hardware troubleshooting, virtualization, containerization, and system performance analysis.

WORK EXPERIENCE

Local Lead | April 2025 - Present
NASA Spaceapps Challenge

- Appointed to local leadership position by NASA following outstanding performance evaluation
- Managing hackathon organization and local team coordination
- Planning and executing international-standard events

Brand and Marketing Lead | March 2024 - October 2024
NASA Spaceapps Challenge

- Developed and implemented social media strategies
- Managed communication channels and increased brand awareness
- Coordinated volunteer teams and event promotion

Computer Hardware Technician 2014 - Present (Part-Time)
Sancak Computer Ltd.

- Computer hardware assembly, maintenance, and repair
- Customer technical support and troubleshooting
- Inventory management and technical consulting

ACADEMIC HISTORY

Istanbul University
Computer Programming

- 1st Grade
- Ongoing

42 Turkiye
Computer Science

- Milestone 3
- Ongoing

CERTIFICATION

- **UAV-1 Sport/Amateur Pilot License** - Turkish Civil Aviation Authority (7553789)
- **Network Security** - Cyber Anatolia Communities (2025)
- **Low Level Hacking and Binary Exploiting** - Cyber Clubs Union (2024)
- **Galactic Team Member** - NASA (2024)
- **CyberStart** - IBM (2024)
- **Cybersecurity Fundamentals** - IBM (2024)
- **Working in Digital World: Professional Skills** - IBM (2024)
- **Explore Emerging Tech** - IBM (2024)
- **Pentesting 101** - BTK Academy (2024)
- **Certificate of Authenticity** - Sui Community (2025)
- **Linux in Cyber Security** - BTK Academy (2024)
- **Creating and Managing Team** - BTK Academy (2024)
- **Low-Level Hacking and Kernel Based Binary Bxploiting** - Cyber Clubs Community(2024)

ACHIEVEMENTS & AWARDS

- Selected as **top performer** in NASA Space Apps Challenge evaluation (2024)
- Appointed as **Local Lead** by NASA (2025)
- Competed as **Mechanical Team Member** in TEKNOFEST Mavi Vatan (2026)
- Earned **The Best Community Impact Award** in Sui Move Hackathon by Sui x 42Turkiye (2025)

TECHNICAL SKILLS

Programming Languages
- C, Go, Python, Rust, PHP, Shell Scripting, Nix Scripting

System
- Red Hat, Debian, Arch, NixOS, Configuration and Management

Virtualization and ContainerTechnologies
- Docker, Virtualbox, Hypervisor, QEMU, Kubernetes

System Programming
- Memory Management, Process Communication, Multi-threading

Networking
- Network Monitoring, Network Security, Network Mapping

Tools & Technologies
- Git, GitHub, Linux, Docker

Cybersecurity
- Penetration Testing, Binary Exploitation, Security Analysis

Libraries/APIs
- Libx, Mutex, pthread, SIGNAL, va_arg, readline, getcwd, sys/wait, LibX

PROJECTS

- **GoNeMap** - Network mapper and monitoring application developed in Go
- **so_long** - 2D game developed in C using Libx library
- **push_swap** - Radix sorting algorithm implementation with two buffers (C)
- **Philosophers** - Dining Philosophers Problem solution using Mutex and pthread (C)
- **libft** - Advanced standard C library clone
- **Minitalk** - Client-server communication project with SIGNAL and geometric growth malloc analysis
- **printf** - Compact printf implementation using va_arg (C)
- **get_next_line** - Line reading and processing function (C)
- **configuration.nix** - Custom NixOS configuration optimized for AMD processors, tailored to maximize system performance for low-level development environments. Includes fine-tuned kernel parameters, resource management, and developer-centric tooling for a seamless, high-efficiency workflow.
- **Minishell**: Bash-like shell. Implemented cd, export, env, ls, echo (with -n flag), cat, cd, piping, input redirection, output redirection, double and sing quote parsing, token lexing and execv commands. Written entirely in C with no memory leaks (including "still reachable").
- **Jetson_Cam_Driver_Dev**: Sony IMX678 camera driver for NVIDIA Jetson Orin Nano, with kernel-level MIPI CSI-2 integration and autofocus support. Includes device-tree overlay and userspace autofocus tooling, supporting resolutions up to 4K30.

LANGUAGES

- **Turkish**: Native
- **English**: Advanced
- **French** : Beginner
- **Arabic** : Beginner